National Institute of Technology Patna

Department of Mathematics

Mid Semester Examination : Oct 2024

MA14102 : Engineering Mathematics I

Branch: CSE I (1st-Semester) **Maximum Marks:** 30

Time: 02.00 hours

Answer all questions

1. For what value of λ, the equations

$$\begin{aligned} x + y + z &= 1 \\ x + 2y + 4z &= \lambda \\ x + 4y + 10z &= \lambda^2 \end{aligned}$$

have a solution. Solve them completely in each case. [5 Marks]

2. Find the inverse of the matrix, using Gauss-Jordan method: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^2 & a & 1 & 0 \\ a^3 & a^2 & a & 1 \end{bmatrix}$. [5 Marks]

3. Find the determinant of the following matrix, $\begin{bmatrix} 3 & 0 & 0 & 1 \\ 0 & 2 & 0 & 5 \\ 6 & -7 & 1 & 0 \\ 3 & 2 & 0 & 6 \end{bmatrix}$. [5 Marks]

4. Show that following vectors in $P_2[x]$ are linearly independent, $p_1(x) = x^2 - 2x + 3$, $p_2(x) = 2x^2 + x + 8$ and $p_3(x) = x^2 + 8x + 7$. [5 Marks]

5. For the matrix given below, determine the four fundamental subspaces generated by the matrix. Further, find the dimensions of all the vector subspaces so obtain, $\begin{bmatrix} 1 & 1 & 1 & -2 \\ 1 & 2 & -1 & 1 \\ 1 & -2 & 7 & -11 \end{bmatrix}$.

[5 Marks]

6. Determine a basis and dimension of $\mathbb{W} = \{[x, y, z, w]^t \in \mathbb{R}^4 \mid x + y - z + w = 0\}$. Also extend the basis to form the basis for $\mathbb{R}^4$. [5 Marks]

*****ALL THE BEST******